

Ana Stringer

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Education

Ph.D. Earth & Climate Sciences, Duke University
National Science Foundation Graduate Research Fellow.
2024 - Present

M.M.A Marine Affairs, University of Washington
Graduate Research Assistant.
2022 - 2024

B.S. Environmental Science & Environmental Sociology, Vanderbilt University.
2018 - 2022

Research Experience

Doctoral Research Assistant, Duke University 2024 - Present
Cryo-HydrO Observation Leaders (COOL) Lab

- Developed and trained a U-Net to automate landfast ice edge detection from Sentinel-2 imagery, producing a time series of ice edge extent for 70 Arctic coastal communities.
- Analyzing freeze-up mechanisms from these outputs by relating landfast extent/edge dynamics to bathymetry and coastline complexity.
- Conducting ethnographic interviews with Arctic communities to document travel access constraints across seasonal trail networks, converting observations into agent-based model rules that use satellite-derived environmental conditions and GCM-based projections to simulate future route accessibility.

Graduate Research Assistant, University of Washington 2022 - 2024
Cryo-HydrO Observation Leaders (COOL) Lab

- Applied ethnographic interview techniques to understand impacts of sea ice reduction on subsistence access.
- Modeled future marine conditions using CESM2 data to understand future impacts.

Publications & Presentations

Ford, J.D., Clerici, S., **Stringer, A.**, Banuet-Martinez, M., Minor, K., Malik, I., Biesbroek, R., Cooley, S., Harper, S., Lowe, J.A., Malleson, N., Ryan, J., Sandiford, S., Woolway, R.I., & Cheung, W.W.L. *Redesigning climate risk modelling from the bottom-up: Creating ethnoclimatologies of climate risk*. **Nature Climate Change (Perspectives)**, under review, February 2025.

St. Laurent, K. A., Beck, T. R., Yacano, M. R., **Stringer, A.**, et al. *A National Assessment of Belowground Decomposition Rates in Tidal Wetlands Using the Tea Bag Index*. **Estuaries and Coasts**, under review (submitted Jan 2025).

Stringer, A. *Automated detection of landfast ice and nearshore polynyas at community-relevant scales*. Oral presentation, **AGU Fall Meeting**, Remote Sensing of the Cryosphere session, New Orleans, December 2025.

Professional Experience

Summer Intern, North Slope Borough

2019 - 2022

Department of Wildlife Management

- Created sea ice use maps derived from interviews with community members and satellite data analysis.
- Developed the North Slope Borough's inaugural marine debris mitigation program successfully securing a \$200k grant from NOAA for implementation.

Awards

National Science Foundation Graduate Research Fellowship (NSF GRFP)

2024

Awarded the prestigious and competitive NSF GRFP, which recognizes and supports exceptional graduate students across the United States.

Program on Climate Change (PCC) Climate Solutions Grant

2024

\$10,000 research grant to support climate-related graduate research at the University of Washington.

Skills

- **Geospatial & Climate Data:** Python, Google Earth Engine, ArcGIS, ICESat-2, Sentinel-1/2, CESM2
- **Machine Learning:** segmentation, clustering, gap-filling, trend analysis
- **Modeling & Analytics:** agent-based modeling, segmentation, clustering, gap-filling, trend analysis
- **Visualization & Reproducibility:** Matplotlib, GIS mapping, GitHub workflows
- **Applied Research:** climate risk indicators, stakeholder engagement, community-centered research